

A. Dedicated Memory in FPGAs

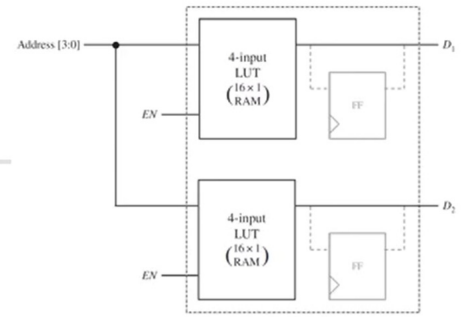
- Memory required in applications designed using FPGA:
 - for storing a *table of constants* to be used as *coefficients* during processing
 - for implementing *instruction* and *data memories* for an embedded processor designed using FPGA
- Early FPGAs:
 - did not contain any dedicated memory
 - Interfaced the FPGAs to *external memory* chips
- Modern FPGAs:
 - incorporate *dedicated memory* on FPGA chips.

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B. LUT/Distributed Memory

- Creating memory from *LUTs* for **LUT-based FPGAs**:
 - If only small amounts of memory are required, we may create that memory using the bits in the *LUTs* w/o using the dedicated memory.
 - E.g.: A 4-variable LUT contains 16 bits of storage.
 - One can create small amounts of memory by combining the storage cells from the LUTs.
 - E.g.: Two 4-input LUTs can be used to create a 32×1 memory or a 16×2 memory.

Creating Memory from LUTs



■ LUT memory:

- memory created from LUT
- called **distributed memory** in Xilinx terminology
- This memory is distributed throughout the chip inside the **logic blocks**.
- Disadv.: Once the LUT memory is used in this fashion, the logic block is generally unusable.

■ The LUT memory can be used as **asynchronous memory**; it can also be combined w/ the logic block **flip-flops** to create **synchronous** memory.

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C. Verilog Models for Inferring Memory in FPGAs

■ Verilog models for inferring memory:

- Embedded memory on FPGAs can be instantiated using **behavioral** Verilog models.
- Memories can be **synchronous** or **asynchronous**:
 - **Async read**: data from the addressed location is available on the output bus after the access time, irrespective of the clock
 - **Distributed memory** is ideal for async memory, since the LUT generates its output asynchronously.
 - **Synch memory read and write**: control lines will have an impact only if the clock is active
- In some memories, **write** is **sync** and **read** is **async**.



Brief Summary

- In modern FPGA synthesis tools, it is possible to indicate whether ***LUT-based RAM*** or ***Block RAM*** is desired via **menu options** in the tools.
- ***Sync reads/writes*** can be obtained with either **block RAM** or **LUT RAM**, but ***async reads*** can be obtained only with **LUT RAM**.
- One should always check the synthesis reports and verify that the desired type of memory was actually created, rather than relying on menu options and warning messages.